

$$1. dX_t = \tau d(X_t + t)dt + \sigma(X_t, t) dB_t \quad \tau u_1 = \tau(X_t, t), \quad \sigma_t = \sigma(X_t, t) \\ (dX_t)^2 = dt, \quad \sigma(B_t) dB_t = 0, (dt)^2 = 0 \\ (dX_t)^2 = (u_1 dt + \sigma_1 dB_t)^2 = u_1^2 (dt)^2 + 2u_1 \sigma_1 dt dB_t + \sigma_1^2 dB_t^2 \\ = \sigma_1^2 dt, \quad (dX_t)^2 = \sigma_1^2 dt (u_1 dt + \sigma_1 dB_t) = \sigma_1^2 u_1 (dt)^2 + \sigma_1^2 dB_t^2 dt = 0$$

$$(a) d(X_t^3) = (X_t + dX_t)^3 - X_t^3 = X_t^3 + 3X_t^2 dX_t + 3X_t dX_t^2 + (dX_t)^3 \\ = X_t^3 + 3X_t^2 dX_t + 3X_t dX_t^2 + (dX_t)^3 = 3X_t^2 dX_t + 3X_t \sigma_1^2 dt + \sigma_1^3 dt \\ = 3X_t^2 dX_t + 3X_t \sigma_1^2 dt + \sigma_1^3 dt = 3X_t^2 (u_1 dt + \sigma_1 dB_t) + 3X_t \sigma_1^2 dt + \sigma_1^3 dt = 3X_t^2 u_1 dt + 3X_t \sigma_1^2 dB_t + 3X_t \sigma_1^2 dt + \sigma_1^3 dt$$

$$(b) f(X_t + dX_t) = f(X_t) + f'(X_t)dX_t + \frac{1}{2} f''(X_t)(dX_t)^2 + \dots$$

$$\therefore df(X_t) = f'(X_t)dX_t + \frac{1}{2} f''(X_t)(dX_t)^2$$

$$f(X_t) = \sin(X_t) \quad \therefore f'(X_t) = \cos(X_t) \quad f''(X_t) = -\sin(X_t)$$

$$\therefore d(\sin(X_t)) = \cos(X_t)dX_t - \frac{1}{2} \sin(X_t)(dX_t)^2$$

$$\therefore dX_t = u_1 dt + \sigma_1 dB_t, \quad (dX_t)^2 = \sigma_1^2 dt$$

$$\therefore d(\sin(X_t)) = \cos(X_t)(u_1 dt + \sigma_1 dB_t) - \frac{1}{2} \sin(X_t) \sigma_1^2 dt$$

$$= (\cos(X_t)u_1 - \frac{1}{2} \sigma_1^2 \sin(X_t)) dt + \sigma_1 \cos(X_t) dB_t$$

$$(c) f(X_t) = e^{X_t} \quad \therefore f'(X_t) = e^{X_t}, \quad f''(X_t) = e^{X_t}$$

$$d(e^{X_t}) = e^{X_t} dX_t + \frac{1}{2} e^{X_t} (dX_t)^2 = e^{X_t} (u_1 dt + \sigma_1 dB_t) + \frac{1}{2} e^{X_t} \sigma_1^2 dt$$

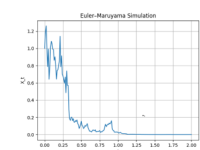
$$= e^{X_t} (u_1 dt + \sigma_1 dB_t) + \frac{1}{2} e^{X_t} \sigma_1^2 dt = (u_1 e^{X_t} + \frac{1}{2} \sigma_1^2 e^{X_t}) dt + \sigma_1 e^{X_t} dB_t$$

$$2. (a) \text{ The SDE can be written in the form } dX_t = aX_t dt + bX_t dB_t$$

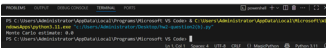
$$\text{where } a=2, b=2. \text{ The Euler-Maruyama discretization is}$$

$$X_{n+1} = X_n + aX_n \Delta t + bX_n \Delta B_n \quad \text{where } \Delta B_n \sim N(0, \Delta t)$$

$$\therefore X_{n+1} = X_n (1 + 2\Delta t + 2\Delta B_n)$$



$$(b) 0.0.$$



$$3. (a) P = \begin{pmatrix} \frac{3}{8} & \frac{1}{6} & \frac{1}{24} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{24} \\ \frac{1}{24} & \frac{1}{6} & \frac{1}{24} \end{pmatrix} \quad u_0 = \begin{pmatrix} \frac{3}{8} \\ \frac{1}{6} \\ \frac{1}{24} \end{pmatrix}$$

$$u_1 = u_0 P = \begin{pmatrix} \frac{3}{8} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \\ \frac{1}{6} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \\ \frac{1}{24} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{8} \\ \frac{1}{6} \\ \frac{1}{24} \end{pmatrix}$$

$$u_2 = u_1 P = \begin{pmatrix} \frac{3}{8} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \\ \frac{1}{6} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \\ \frac{1}{24} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{8} \\ \frac{1}{6} \\ \frac{1}{24} \end{pmatrix}$$

$$u_3 = u_2 P = \begin{pmatrix} \frac{3}{8} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \\ \frac{1}{6} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \\ \frac{1}{24} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{24} \times \frac{1}{24} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{8} \\ \frac{1}{6} \\ \frac{1}{24} \end{pmatrix}$$

$$(b) \text{ Suppose } \pi_0 = (\pi_{0,0}, \pi_{0,1}, \pi_{0,2}) \quad \pi_0 + \pi_1 + \pi_2 = 1$$

$$\pi_0 = \pi_0 P \Rightarrow (\pi_{0,0}, \pi_{0,1}, \pi_{0,2}) = (\pi_{0,0}, \pi_{0,1}, \pi_{0,2}) \begin{pmatrix} \frac{3}{8} & \frac{1}{6} & \frac{1}{24} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{24} \\ \frac{1}{24} & \frac{1}{6} & \frac{1}{24} \end{pmatrix}$$

$$\therefore \begin{cases} \pi_{0,0} = \frac{1}{8} \pi_{0,0} + \frac{1}{6} \pi_{0,1} & (1) \\ \pi_{0,1} = \frac{1}{6} \pi_{0,0} + \frac{1}{6} \pi_{0,1} & (2) \\ \pi_{0,2} = \frac{1}{24} \pi_{0,0} + \frac{1}{6} \pi_{0,1} + \frac{1}{24} \pi_{0,2} & (3) \end{cases}$$

$$\text{From (1) we get } \pi_{0,0} = \pi_{0,1}, \text{ from (2) we get } \pi_{0,0} = 2\pi_{0,1}$$

$$\therefore \pi_{0,0} + \pi_{0,1} + \pi_{0,2} = 1 \quad \therefore \pi_{0,0} + \frac{1}{2} \pi_{0,0} + \pi_{0,2} = 1 \quad \therefore 1.5\pi_{0,0} = 1$$

$$\pi_{0,0} = \frac{2}{3} \quad \therefore \pi_{0,1} = \frac{1}{3} \quad \therefore \pi_{0,2} = \frac{1}{3}$$

$$(c) \text{ (a) Based on the definition of lazy random walk, at each step}$$

$$\text{from vertex } i, \text{ the chain stays at } i \text{ with probability } \frac{1}{2}, \text{ the remaining}$$

$$\frac{1}{2} \text{ is equally distributed among all neighbors of } i.$$

$$\therefore P_{ij} = \begin{cases} \frac{1}{2}, & i=j \\ \frac{1}{2 \deg(i)}, & j \text{ is a neighbor of } i \\ 0, & \text{otherwise} \end{cases}$$

$$\text{for graph (i): } \deg(1)=2, \deg(2)=2, \deg(3)=3, \deg(4)=3, \deg(5)=2, \deg(6)=2$$

$$\therefore \text{the transition matrix (ordering vertices } 1, 2, 3, 4, 5, 6) \text{ is}$$

$$P^{(i)} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{6} & \frac{1}{2} & \frac{1}{6} & 0 \\ 0 & 0 & 0 & \frac{1}{6} & \frac{1}{2} & \frac{1}{6} \\ 0 & 0 & 0 & \frac{1}{6} & \frac{1}{6} & \frac{1}{2} \end{pmatrix}$$

$$\text{for graph (ii): } \deg(1)=3, \deg(2)=3, \deg(3)=4, \deg(4)=3, \deg(5)=4, \deg(6)=3$$

$$\therefore \text{the transition matrix (ordering vertices } 1, 2, 3, 4, 5, 6) \text{ is}$$

$$P^{(ii)} = \begin{pmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{6} & 0 & 0 & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{6} & 0 & \frac{1}{6} & 0 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{3} & \frac{1}{6} & \frac{1}{6} & 0 \\ 0 & 0 & \frac{1}{6} & \frac{1}{2} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{2} & \frac{1}{6} \\ \frac{1}{6} & 0 & 0 & \frac{1}{6} & \frac{1}{6} & \frac{1}{2} \end{pmatrix}$$

$$(b) \text{ The eigenvalues were computed numerically using Python.}$$

$$\text{For graph (i): } \{1, 0.8977, 0.4167, 0.25, 0.25, 0.1857\}$$

$$\text{For graph (ii): } \{1, 0.6296, 0.5292, 0.4219, 0.2624, 0.1549\}$$

$$(c) \therefore \text{For a lazy random walk, the largest eigenvalue is always } 1.$$

$$\text{The connectivity of the graph is measured by the spectral gap:}$$

$$1 - \lambda_2$$

$$\text{for graph (i): } \lambda_2 \approx 0.8977 \quad \therefore 1 - \lambda_2 \approx 0.1023$$

$$\text{for graph (ii): } \lambda_2 \approx 0.6296 \quad \therefore 1 - \lambda_2 \approx 0.3704$$

$$\therefore \text{Graph (ii) has a larger spectral gap, the random walk mixes faster.}$$

$$\therefore \text{We can see from the transition matrix eigenvalues:}$$

$$\text{the smaller second-largest eigenvalue indicates stronger connectivity.}$$

$$\therefore \text{Graph (ii) is more connected.}$$

$$5. \text{ Assume } X \text{ is finite and let } n = |X|, \text{ define the uniform}$$

$$\text{distribution } \pi(x) = \frac{1}{n}, \text{ for all } x \in X$$

$$\text{for } \forall y \in X, (\pi P)(y) = \sum_{x \in X} \pi(x) P(x, y) = \frac{1}{n} \sum_{x \in X} P(x, y)$$

$$\therefore P \text{ is symmetric } \therefore \sum_{x \in X} P(x, y) = \sum_{x \in X} P(y, x)$$

$$\therefore P \text{ is transition matrix } \therefore \text{each row sums to } 1$$

$$\therefore \sum_{x \in X} P(y, x) = \sum_{x \in X} P(x, y) = 1 \quad \therefore (\pi P)(y) = \frac{1}{n} = \pi(y)$$

$$\therefore \pi = \pi P \quad \therefore \text{the uniform distribution is stationary.}$$

